1

a

At our α level of 0.05, we achieve the following statistics about the correlation of our two variables:

```
phi obs = -0.1989717932944467 perm p = 0.0044995500449955
jaccard obs = 0.0 perm p = 1.0
chi2 obs = 7.87836513083478 perm p = 0.005099490050994901
```

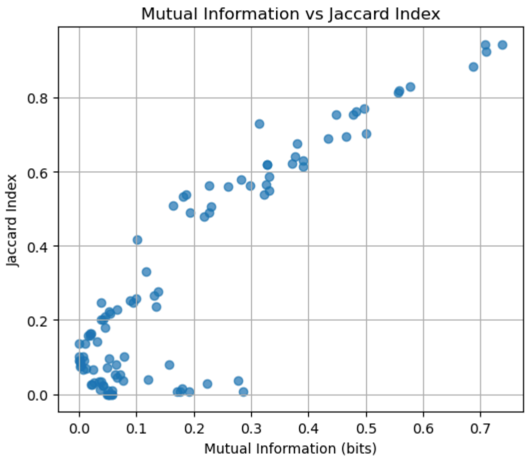
These three statistics show that they all show low correlation with high surity, hence the values close to 0 for all three statistics. All of our p values are extreme as well, showing that our tests are pretty sure that the uncorrelation is accurate.

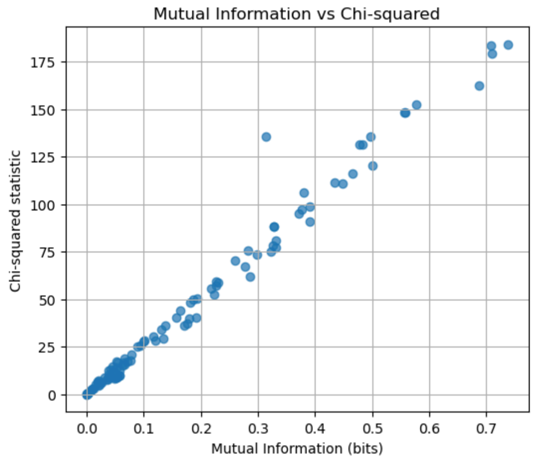
b

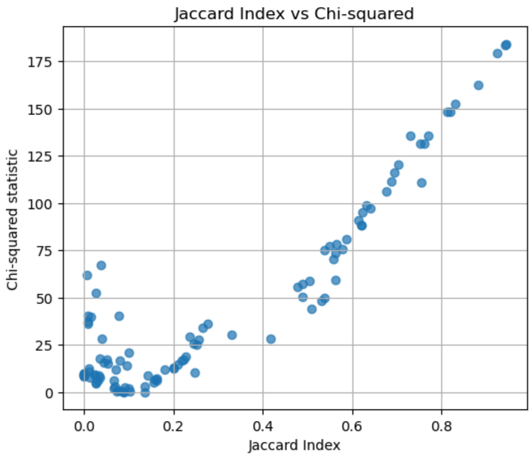
This is an example of the correlations of the columns of the data for part b:

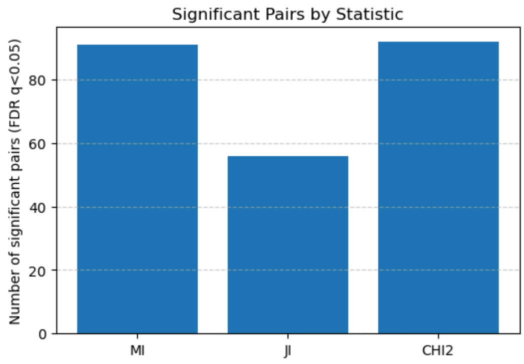
```
        pair   i   j   MI_stat      MI_p   JI_stat      JI_p   CHI2_stat
CHI2_p
0      V1-V2   1   2  0.054093  0.002997  0.000000  1.000000    9.211455
0.003996
1      V1-V3   1   3  0.377032  0.000999  0.641026  0.000999   97.401858
0.000999
2      V1-V4   1   4  0.709891  0.000999  0.925926  0.000999  179.312702
0.000999
3      V1-V5   1   5  0.056741  0.000999  0.000000  1.000000    9.684875
0.001998
4      V1-V6   1   6  0.191292  0.000999  0.007812  1.000000   40.528022
0.000999
..       ...  ..  ..       ...       ...       ...       ...         ...
 ...
100  V12-V14  12  14  0.000053  1.000000  0.089744  0.531469    0.014794
1.000000
101  V12-V15  12  15  0.019314  0.023976  0.163265  0.016983    6.340686
0.020979
102  V13-V14  13  14  0.079063  0.000999  0.100000  1.000000   20.765972
0.000999
103  V13-V15  13  15  0.038066  0.001998  0.247525  0.002997   10.406218
0.004995
```

```
104  V14-V15  14  15  0.000100  1.000000  0.136364  0.514486   0.027695
1.000000
```





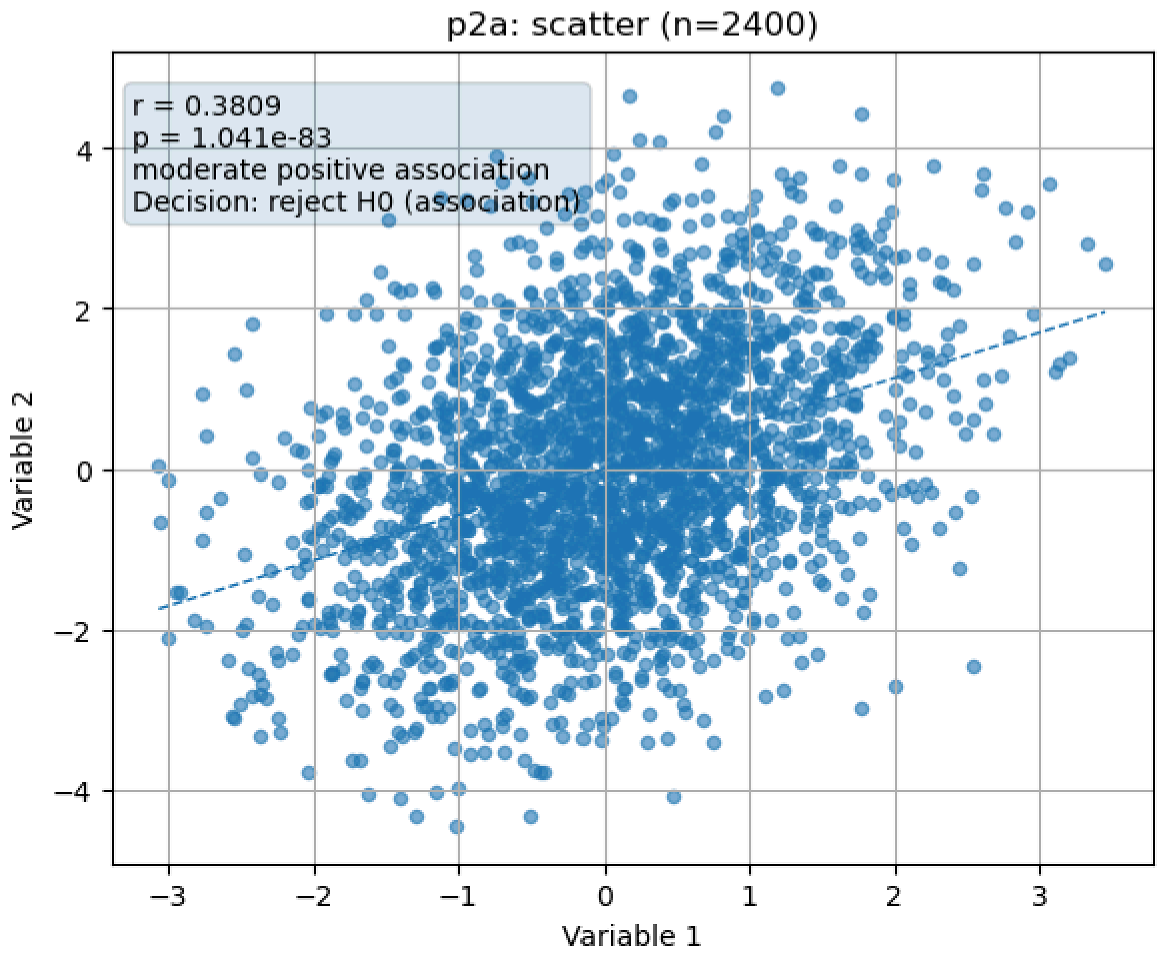




We can see from these plots that the statistics are fairly well correlated with each other, particularly mutual information and chi squared. While the jaccard index is fairly correlated with the other two, it is non-linear toward the bottom end. I personally trust and like to utilize the chi-squared test as the distribution is easier for me to understand and comprehend.

2

a



As seen in the plot for part A, there is not much correlation between the variables. With our p value significantly below 0.05, we can confidently say that these variables do not have much association.

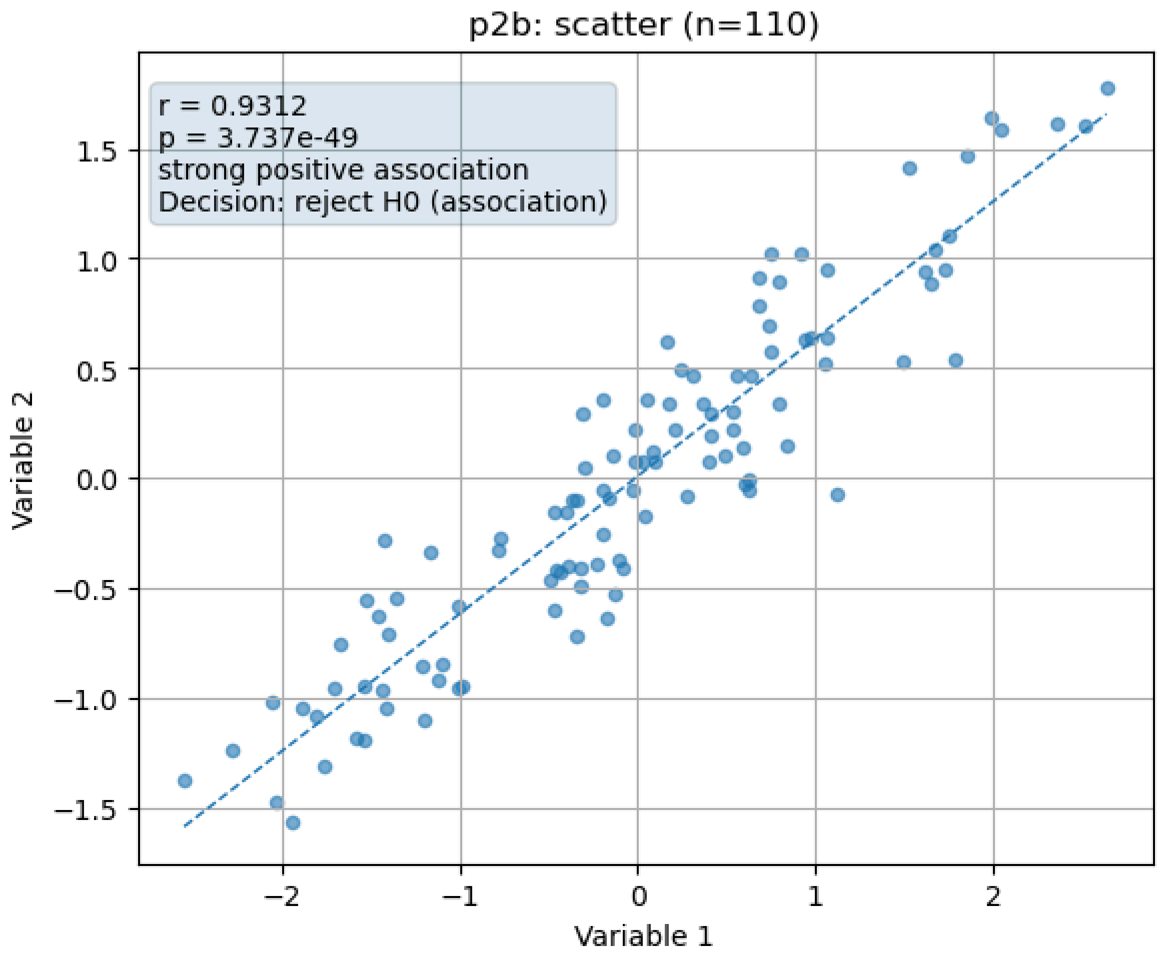
b

```
p2a: n=2400, r=0.3809, p=1.041e-83 → reject H0 (association)
p2b: n=110, r=0.9312, p=3.737e-49 → reject H0 (association)

By correlation magnitude: p2b shows stronger association (|r| 0.9312 >
0.3809).
By p-value: p2a shows stronger evidence against H0 (smaller p-value:
1.041e-83 < 3.737e-49).
```

We can see by our analysis here that by comparison with the data from part A, B theoretically has a higher correlation than A if you are looking at correlation, and A higer than B if looking at probability.

This discrepancy comes from the fact that the A dataset is significantly larger than that of the B dataset, making its probabilities much more sure.



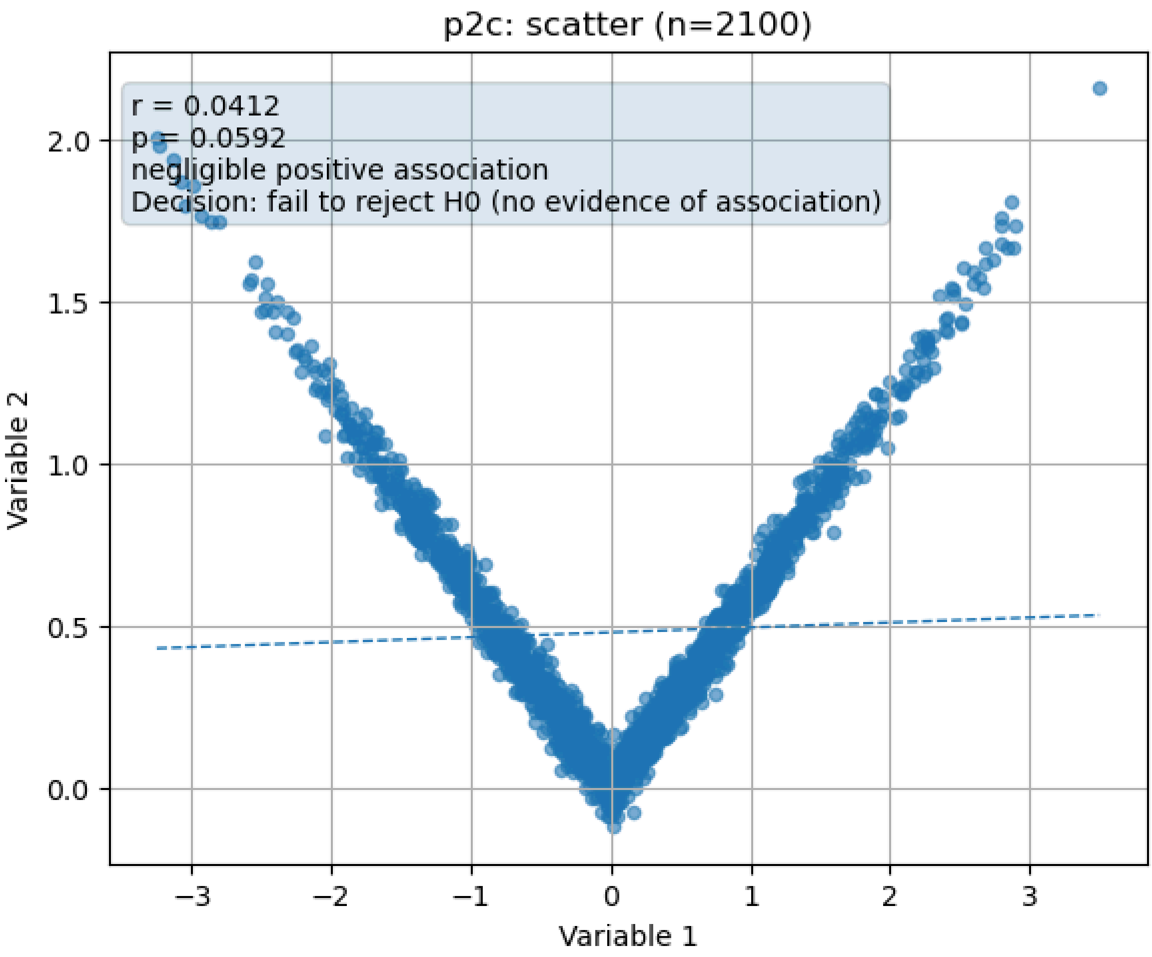
I would say that the data from part B would be significantly stronger than that of A. I would interpret this as high correlation, low surity, but part A has moderate correlation with high surity. So if you are asking for correlation, B is much stronger.

C

```
p2a: n=2400, r=0.3809, p=1.041e-83 → reject H0 (association)
p2c: n=2100, r=0.0412, p=0.0592 → fail to reject H0 (no evidence of
association)

By correlation magnitude: p2a shows stronger association (|r| 0.3809 >
0.0412).
By p-value: p2a shows stronger evidence against H0 (smaller p-value:
1.041e-83 < 0.0592).
```

By the numbers this time, A has a higher correlation than C, while A has a much smaller p-value. I would say this p value is also for the similar reason to last time, A having a linear-ish shape and many data points leading to a lower probability. While C has almost as many points, its points are much more consistent, leading to less exploration of the valid area, leading to a higher p value.



This time however, C does not have a linear correlation so this test completely fails to evaluate how well they are correlated. C clearly has a correlation that is non-linear and the test predicts that there is little to no correlation. In this case, I would call C significantly more correlated than A, although not linearly.